Complete Week 1 to Week 52 Software Engineering Roadmap

Step-by-Step Self-Taught Study Plan from Zero Coding Experience to Remote Full-Stack Engineer

How to Execute Every Week From Home (15–20 Hours/Week)

This master plan breaks your self-taught journey into **52 consecutive weekly sprints**. Instead of passively watching videos, treat every week like a miniature software engineering sprint with a concrete piece of working software pushed to GitHub by Sunday evening.

- **Monday – Friday (2 hours/day):** Spend the first 30 minutes reading official documentation or completing the week's core module, the next 80 minutes actively coding and debugging in Visual Studio Code, and the final 10 minutes committing your code to GitHub with a 3-sentence summary of what you built and fixed.
- **Saturday (3.5 hours — Deep Build Block):** Build the week's listed *Deliverable & Milestone* from scratch without following a video step-by-step. Use official documentation (MDN Web Docs, React Docs, Node.js Learn) when you get stuck.
- **Sunday (1.5 hours — Review, Refactor & Deploy):** Clean up variable names, write a clear `README.md` explaining what you built, deploy your work to a live URL, and complete 2 coding exercises on the Exercism JavaScript Track.

Stage 1: Computer Science, Terminal & Git Foundations (Weeks 1–6)

Your first six weeks establish how computers execute logic, how to operate your computer from the command line, and how professional engineers track code changes using Git and GitHub.

Week	Core Topic & Primary Resource	Step-by-Step Weekly Actions	Weekly Deliverable & Milestone
Week 1	Computational Thinking & Local Setup Harvard CS50x (Week 0) & VS Code Guide	Watch CS50x Lecture 0 on binary, algorithms, and abstraction. Install Visual Studio Code, configure extensions, and learn keyboard shortcuts.	Complete CS50x Problem Set 0 interactive logic project and configure your local developer workspace.
Week 2	Variables, Conditionals & Loops Harvard CS50x (Week 1) or edX CS50	Study data types, boolean logic, `if/else` branching, `for/while` loops, and command-line compilation.	Write and test command-line programs that generate Mario-style ASCII pyramids and calculate greedy coin change.
Week 3	Arrays, Strings, Functions & Debugging Harvard CS50x (Weeks 2–3)	Learn how arrays and strings sit in memory, step through code line-by-line with a debugger, and compare linear vs. binary search.	Build a command-line Caesar/substitution cipher tool and a plurality voting tally program.

Week	Core Topic & Primary Resource	Step-by-Step Weekly Actions	Weekly Deliverable & Milestone
Week 4	How the Internet Works & Command Line MDN How the Web Works & The Odin Project	Study DNS, HTTP requests/responses, and browser rendering. Master Unix/Linux terminal commands (`cd`, `ls`, `mkdir`, `mv`, `rm`, `grep`).	Create a local project folder hierarchy and automate file creation entirely via terminal commands without using a mouse.
Week 5	Git Version Control & GitHub Workflow GitHub Skills & Git Roadmap	Configure SSH keys, learn `git init`, `add`, `commit`, `branch`, `checkout`, `merge`, and practice resolving merge conflicts on GitHub.	Complete the "Introduction to GitHub" and "Communicate using Markdown" courses on GitHub Skills.
Week 6	Stage 1 Checkpoint & Static Hosting GitHub Pages Quickstart & MDN First Website	Connect your local Git workflow to live web hosting. Review how browsers fetch HTML files from a remote server.	Publish a live "Learning Log" website at `username.github.io` via GitHub Pages documenting your Stage 1 code.

Stage 2A: Semantic HTML5, Modern CSS & Responsive Design (Weeks 7–12)

In weeks 7 through 12, you master the visual and structural layer of the web so you can translate any design mockup into a clean, accessible, mobile-friendly webpage without relying on pre-made templates.

Week	Core Topic & Primary Resource	Step-by-Step Weekly Actions	Weekly Deliverable & Milestone
Week 7	Semantic HTML5, Forms & Accessibility MDN Structuring Content & Odin Foundations	Learn semantic landmarks (`header`, `nav`, `main`, `article`, `section`), accessible form inputs, labels, tables, and screen-reader attributes.	Build and deploy a multi-page Recipe & Technical Documentation site with validated HTML5 forms.
Week 8	CSS Cascade, Box Model & Typography MDN What is CSS? & MDN Styling Content	Master CSS specificity, inheritance, `box-sizing: border-box`, margin/padding/border mechanics, relative units (`rem`, `em`, `vh`/`vw`), and custom CSS variables.	Transform your Week 7 HTML site with a cohesive color palette, modular type scale, and card layouts.
Week 9	One-Dimensional Layouts with Flexbox The Odin Project Foundations (Flexbox Module)	Study main vs. cross axes, `justify-content`, `align-items`, `flex-wrap`, and `flex: 1` growth/shrink behavior across nested containers.	Build a pixel-accurate SaaS Product Landing Page with a sticky navigation bar, hero section, feature rows, and pricing cards.
Week 10	Two-Dimensional Layouts with CSS Grid freeCodeCamp Full-Stack Curriculum	Learn `grid-template-columns`, `minmax()`, `auto-fit`, `grid-template-areas`, and when to combine CSS Grid for macro page layout with Flexbox for micro components.	Build a complete SaaS Admin Dashboard UI featuring a collapsible sidebar, header toolbar, project grid, and activity feed.

Week	Core Topic & Primary Resource	Step-by-Step Weekly Actions	Weekly Deliverable & Milestone
Week 11	Responsive Design & Media Queries MDN Learn Web Development	Adopt mobile-first CSS architecture, fluid images (`max-width: 100%`), responsive breakpoints, and CSS transitions/transforms.	Audit and refactor your Landing Page and Admin Dashboard so both work seamlessly at 375px (mobile), 768px (tablet), and 1440px (desktop).
Week 12	Stage 2A Milestone: Portfolio Shell v1 roadmap.sh HTML & GitHub Pages	Combine semantic HTML5, CSS Grid, Flexbox, dark/light theme variables, and keyboard focus styling into an original portfolio shell.	Deploy Personal Developer Portfolio v1 live on GitHub Pages achieving 95+ scores across Performance and Accessibility in Chrome Lighthouse.

Stage 2B: Core JavaScript, DOM Manipulation & Async APIs (Weeks 13–18)

Before touching frontend frameworks, spend six weeks mastering core JavaScript so you can manipulate webpage elements, manage application data structures, and fetch live data from external servers.

Week	Core Topic & Primary Resource	Step-by-Step Weekly Actions	Weekly Deliverable & Milestone
Week 13	JavaScript Fundamentals & Functions MDN Dynamic Scripting & Odin Foundations	Learn `let`/`const`, arrow functions, scope, closures, template literals, and Chrome DevTools console debugging.	Build a browser-based Rock-Paper-Scissors game with score tracking and solve your first 5 exercises on Exercism JavaScript.
Week 14	DOM Manipulation & Event Handling The Odin Project Foundations	Master `querySelector`, dynamic element creation, event delegation, keyboard/mouse listeners, and form submission handling.	Build an interactive Etch-a-Sketch Pixel Art Pad with adjustable grid dimensions, color modes, and clear controls.
Week 15	Arrays, Objects & Functional Methods Exercism JavaScript Track & MDN Scripting	Master array/object destructuring, spread/rest operators, and array transformations (`map`, `filter`, `reduce`, `find`, `some`, `every`).	Build a full On-Screen Calculator supporting chained operations, decimals, backspace, and full keyboard input without using `eval()`.
Week 16	Object-Oriented JS, Modules & Storage Odin Full Stack JavaScript Path	Learn ES6 classes, factory functions, module imports/exports, JSON serialization, and persisting state in browser `localStorage`.	Build a multi-project To-Do & Habit Tracker with priority filtering, due dates, and automatic `localStorage` persistence.
Week 17	Asynchronous JS: Promises & `fetch` MDN Dynamic Scripting & Odin JS Path	Understand the JavaScript event loop, Promises, `async`/`await`, `try`/`catch` error states, loading spinners, and working with REST JSON APIs.	Build a live Weather & Air Quality Dashboard that fetches real-time meteorological data by city search with unit toggling (°C/°F).

Week	Core Topic & Primary Resource	Step-by-Step Weekly Actions	Weekly Deliverable & Milestone
Week 18	Stage 2B Milestone: Vanilla JS Capstone Frontend Beginner Roadmap	Synthesize DOM architecture, modular ES6 state management, API fetching, and responsive CSS without any external UI libraries.	Ship an interactive Kanban Task Board with column movement, search/tag filtering, and clean Git commit history.

Stage 3A: Modern Frontend Engineering with React & TypeScript (Weeks 19–26)

Modern remote engineering teams build user interfaces using component-based frameworks and static typing. In weeks 19 through 26, you transition from Vanilla JavaScript to production-grade React, TypeScript, and automated frontend testing.

Week	Core Topic & Primary Resource	Step-by-Step Weekly Actions	Weekly Deliverable & Milestone
Week 19	Modern Tooling (Vite) & React Components React Quick Start & Thinking in React	Learn npm package management, Vite bundling, JSX syntax, declarative UI rendering, component composition, and passing data via props.	Migrate your Portfolio and Kanban UI into reusable, modular React functional components.
Week 20	React State (`useState`) & Controlled Forms Odin Full Stack JS Path (React Module)	Master state immutability, lifting state up between sibling components, controlled form inputs, and dynamic list rendering with keys.	Build an interactive Resume/CV Builder app where users edit form sections on the left and export a live formatted preview on the right.
Week 21	Side Effects (`useEffect`) & Custom Hooks University of Helsinki Full Stack Open	Learn how to synchronize components with external APIs, clean up event listeners/timers, and extract reusable data-fetching hooks.	Build a PokeAPI/GitHub API Memory Card & Explorer app with custom data-fetching hooks, caching, and score persistence.
Week 22	Client-Side Routing & Global State React Official Reference & Odin React	Implement multi-page SPA navigation with React Router, dynamic URL parameters, protected routes, and shared state via React Context.	Build a multi-page E-Commerce Storefront with product category filtering, dynamic item pages, and a slide-out shopping cart.
Week 23	TypeScript Fundamentals for React TypeScript Handbook & Basic Types	Learn static type annotations, interfaces, union types, generics, type narrowing, and typing React props, state, and DOM events.	Refactor your E-Commerce Storefront from JavaScript to strict TypeScript (`strict: true`) with zero any types.
Week 24	Automated Frontend Unit & UI Testing Full Stack Open (Testing Module)	Learn Test-Driven Development (TDD) concepts, component rendering tests, mocking API responses, and asserting user interactions.	Write an automated test suite covering 80%+ of your E-Commerce cart math, discount logic, and form validations.

Week	Core Topic & Primary Resource	Step-by-Step Weekly Actions	Weekly Deliverable & Milestone
Week 25	Full-Stack React & Next.js Architecture Full Stack Open: Next.js & Creating a React App	Study Server Components vs. Client Components, file-based routing, server actions, and SEO/performance optimization.	Build a fast, server-rendered Developer Blog & Project Showcase using Next.js and TypeScript.
Week 26	Stage 3A Milestone: Frontend Capstone roadmap.sh Frontend Roadmap	Combine TypeScript, React Router/Next.js, accessible UI components, data visualization, and automated tests into a polished web app.	Deploy a production Personal Finance & Budget Analytics Dashboard with interactive charts, CSV export, and automated tests.

Stage 3B: Backend Engineering with Node.js, Express & PostgreSQL (Weeks 27–34)

To become a self-sufficient full-stack engineer, weeks 27 through 34 teach you how to build secure backend servers, design relational SQL databases, authenticate users, and deploy full-stack APIs.

Week	Core Topic & Primary Resource	Step-by-Step Weekly Actions	Weekly Deliverable & Milestone
Week 27	Node.js Runtime, Modules & File System Node.js Learn & Node.js Setup	Understand server-side JavaScript execution, environment variables, the Node.js event loop, asynchronous file I/O, and raw HTTP servers.	Build a multi-route Node.js server and a CLI tool that converts a folder of Markdown notes into styled HTML pages.
Week 28	Express.js Framework & RESTful APIs Express Starter Guide & MDN Node/Express	Master Express routing, controllers, request/response lifecycle, custom error-handling middleware, and input validation.	Build a RESTful Inventory Management API supporting full CRUD (`GET`, `POST`, `PUT`, `DELETE`) for products and categories.
Week 29	Relational Databases & PostgreSQL SQL PostgreSQL Official Tutorial & Neon SQL Guide	Learn relational schema design, primary/foreign keys, normalization, indexes, transactions, and complex `JOIN` and `GROUP BY` queries.	Design a multi-table SQL schema for an online marketplace and write 20 analytical SQL queries against sample data.
Week 30	Database Integration & ORM/Query Builders PostgreSQL Docs & Odin NodeJS Path	Connect Express to PostgreSQL using parameterized queries (preventing SQL injection) and manage schema migrations with Prisma or node-pg.	Wire your Week 28 Inventory API to a live PostgreSQL database with automated seed scripts and pagination.
Week 31	User Authentication, Sessions & JWTs Odin Full Stack JS Path (Auth Module)	Learn password hashing with `bcrypt`, session cookies vs. JSON Web Tokens (JWT), OAuth flows, and role-based authorization middleware.	Build a "Members-Only" Community Clubhouse app where verified members see author names and admins moderate posts.

Week	Core Topic & Primary Resource	Step-by-Step Weekly Actions	Weekly Deliverable & Milestone
Week 32	API Security, File Uploads & Caching Backend Developer Roadmap	Implement rate limiting, CORS policies, Helmet security headers, multipart file uploads to cloud storage, and background email/job queues.	Build a Cloud Drive File Manager app allowing authenticated users to create folders, upload files, and generate shareable links.
Week 33	Backend Testing, CI/CD & Docker Basics GitHub Learn & Node.js Debugging	Write integration tests against a test PostgreSQL database, containerize your app with a `Dockerfile`, and set up GitHub Actions CI.	Configure a GitHub Actions pipeline that automatically lints, builds, and runs API integration tests on every pull request.
Week 34	Stage 3B Milestone: Full-Stack Blog Platform Full Stack Open & Full Stack Roadmap	Connect a React + TypeScript frontend client to your authenticated Express + PostgreSQL REST API across separate environments.	Deploy a complete Full-Stack Publishing Platform featuring a public reader site, nested comments, and a private author CMS dashboard.

Stage 4A: Production Capstone Portfolio & Open Source (Weeks 35–42)

Remote employers hire self-taught engineers who prove they can build original, production-grade software and collaborate asynchronously on existing codebases. Dedicate weeks 35 through 42 to building two flagship capstones and contributing to open source.

Week	Core Topic & Primary Resource	Step-by-Step Weekly Actions	Weekly Deliverable & Milestone
Week 35	Capstone #1 Specification & DB Architecture Full Stack Career Guide	Scope an original B2B SaaS application (e.g., an Async Remote Team Standup, Issue Tracker, or Client Portal). Draft user stories, API specs, and ER diagrams.	Publish a technical specification document, PostgreSQL database schema, and Figma/wireframe layout in your Capstone #1 repo.
Week 36	Capstone #1 Backend API & Auth Engine Express.js & PostgreSQL	Build the complete backend service with TypeScript, PostgreSQL migrations, JWT/cookie auth, role permissions (Owner, Admin, Member), and seed data.	Complete and test 100% of Capstone #1's REST API endpoints with automated integration tests passing in GitHub Actions.
Week 37	Capstone #1 React UI & State Integration React Learn & TypeScript Docs	Build the responsive frontend client with optimistic UI updates, skeleton loading states, keyboard shortcuts, search/pagination, and error boundaries.	Connect the React + TypeScript frontend to your live backend API and include a 1-click "Demo Account Login" button for recruiters.
Week 38	Capstone #1 Launch, Docs & Async Video Demo GitHub Skills	Deploy frontend and backend to production, write a comprehensive `README.md` (architecture diagram, trade-offs, setup), and record a 2-minute video demo.	Ship Capstone #1 live with a clickable demo link, 1-click recruiter login, and embedded 2-minute architecture walkthrough video.

Week	Core Topic & Primary Resource	Step-by-Step Weekly Actions	Weekly Deliverable & Milestone
Week 39	Capstone #2: Real-Time or Transactional App (Part 1) Odin Full Stack JS (Final Project)	Architect a second flagship app showcasing real-time WebSockets or complex SQL transactions (e.g., Live Collaborative Workspace or Booking Marketplace).	Build the backend real-time messaging/notification engine and relational database layer with automated tests.
Week 40	Capstone #2: Frontend Polish & Live Launch (Part 2) Full Stack Open	Complete the responsive UI, run Lighthouse performance/accessibility audits, write end-to-end tests, and record your second 2-minute video walkthrough.	Deploy Capstone #2 to production with full documentation and a live demo login.
Week 41	Navigating Large Open-Source Codebases freeCodeCamp GitHub & Exercism GitHub	Clone an active open-source project, follow its `CONTRIBUTING.md` setup guide, run its test suite locally, and trace how a feature works across files.	Successfully build and run an established open-source repository locally and claim your first good `first issue`.
Week 42	Submitting Open-Source Pull Requests GitHub Learn & GitHub Resources	Write clean code or documentation patches, open well-formatted pull requests referencing the issue number, and respond constructively to maintainer code reviews.	Have at least 2 open-source pull requests reviewed and merged into public community repositories.

Stage 4B: Interview Prep, Async Proof-of-Work & Remote Job Search (Weeks 43–52)

Your final 10 weeks combine structured coding interview preparation with a high-conversion remote job outreach campaign targeting distributed startups and remote-first engineering teams.

Week	Core Topic & Primary Resource	Step-by-Step Weekly Actions	Weekly Deliverable & Milestone
Week 43	Algorithms I: Arrays, Hashing & Two Pointers NeetCode Roadmap & Exercism JS	Study Big-O time and space complexity. Practice hash map lookups, frequency counters, and two-pointer array manipulation in TypeScript/JavaScript.	Solve and document 15 foundational Array, String, and Two-Pointer problems on NeetCode.
Week 44	Algorithms II: Sliding Window, Stacks & Lists NeetCode	Master sliding window substrings, stack parsing, binary search on sorted arrays, and linked list pointer manipulation while explaining your thought process out loud.	Solve 15 Sliding Window, Stack, and Binary Search problems within 30-minute timed blocks.
Week 45	Algorithms III: Trees, BFS/DFS & Recursion NeetCode Roadmap	Learn recursion call stacks, binary tree depth-first search (DFS), breadth-first search (BFS), and basic graph traversal for technical interviews.	Solve 12 Tree and Graph traversal problems and review practical frontend DOM-tree traversal equivalents.

Week	Core Topic & Primary Resource	Step-by-Step Weekly Actions	Weekly Deliverable & Milestone
Week 46	Final Portfolio, Resume & GitHub Profile Polish The Odin Project (Getting Hired)	Pin your top 4 repositories on GitHub, embed your 2-minute video walkthroughs on your portfolio site, and write a results-driven 1-page engineering resume.	Launch Final Portfolio v2, pinned GitHub profile README, and ATS-ready PDF resume highlighting your capstones and open-source PRs.
Week 47	Remote Startup Outreach & Founder Notes Wellfound Remote Jobs & Junior Startups	Filter for remote roles on Wellfound Full-Stack. For each application, send a concise 4-sentence note to the engineering lead linking a matching project.	Submit 15 tailored remote startup applications with custom product notes and track every outreach in a job-search CRM sheet.
Week 48	Global Remote Job Boards & Async Proof We Work Remotely & Remotive Jobs	Apply daily to WWR Full-Stack, WWR Front-End, and Remotive listings within 48 hours of posting, emphasizing async communication.	Submit 15 high-fit remote applications and publish 1 technical blog article breaking down an architecture challenge from your capstone.
Week 49	Mastering Practical Coding & Take-Home Screens React Docs & Express Docs	Remote startups frequently test candidates with a 3–4 hour practical React/API build or live feature pairing rather than whiteboard riddles.	Complete 2 timed 3-hour mock take-home challenges (building a paginated, searchable API + React UI with tests and a clean README).
Week 50	System Design Basics & Behavioral Stories roadmap.sh Full Stack	Study web architecture fundamentals (load balancers, caching, database indexing, CDNs) and prepare 6 STAR behavioral stories focused on self-management.	Record yourself answering the 10 most common remote engineering interview questions and refine your delivery.
Week 51	Live Mock Interviews & Pipeline Follow-Ups NeetCode & Wellfound Remote	Complete live mock technical interviews, send polite 7-day follow-up messages to hiring managers with a link to a newly shipped feature, and iterate on feedback.	Complete 3 full mock interviews (1 algorithmic, 1 live React/Node feature build, 1 behavioral) and maintain 15 weekly applications.
Week 52	Evaluating Offers & First 90-Day Remote Plan We Work Remotely & Remotive	Compare remote employment structures (direct hire vs. global Employer of Record/contractor), negotiate compensation, and set up your daily async standup workflow.	Sign your first remote software engineering offer (or paid freelance/contract bridge role) and prepare your 30-60-90 day onboarding plan.